
THE UNCONVENTIONAL WEEKLY

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UNCONVENTIONAL NEUROSCIENTISTS



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ALZHEIMER'S REVERSAL USING FDA-APPROVED CANCER DRUGS

Breakthrough alert: FDA-approved cancer drugs show strong promise in reversing Alzheimer's—opening exciting new treatment possibilities and hope for patients.

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What's Happening?

Researchers at UCSF and the Gladstone Institutes discovered that two FDA-approved cancer drugs, letrozole and irinotecan, can reverse gene expression changes linked to Alzheimer's disease. In mouse models, these drugs dissolved toxic tau protein aggregates—a hallmark of Alzheimer's—and restored memory and cognitive function. Because both drugs are already approved for cancer treatment, their safety profiles are well understood, potentially allowing faster transition to human trials.

The researchers used computational drug-repurposing by screening large gene expression datasets to find drugs that counter Alzheimer's-associated molecular changes. This innovative method leverages existing drugs to accelerate treatment discovery, bypassing the lengthy process of developing entirely new compounds.

Why It Matters?

This research shifts Alzheimer's treatment from symptom relief toward addressing disease mechanisms by targeting molecular drivers of neurodegeneration. By clearing tau pathology and reversing gene dysregulation, these drugs have potential to modify disease progression, a major unmet need in Alzheimer's care.

Furthermore, analysis of 1.4 million patient health records found that people prescribed these cancer drugs had a lower risk of Alzheimer's, providing strong real-world evidence supporting their therapeutic potential. This combined laboratory and epidemiological data opens promising avenues for repurposing established medications.

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Why It's Unconventional?

This study takes a bold, outside-the-box approach by looking beyond the traditional “brain-only” focus in Alzheimer’s drug discovery. Instead of searching for brand-new compounds designed specifically for the brain, researchers tapped into the vast arsenal of existing cancer drugs—medications already proven safe in humans—and asked if they could be repurposed based on similarities in gene expression patterns.

What makes this especially innovative is how the team combined computational biology with large-scale patient data. This challenges the idea that Alzheimer’s treatments must be new brain-specific drugs and could speed up finding effective therapies.

What’s your **Unconventional** Take?

- Could this change how we approach treatment for other brain diseases?
- Are we overlooking the potential of existing drugs?

*Do you think this is a breakthrough waiting to happen
or too risky to translate to humans?*

